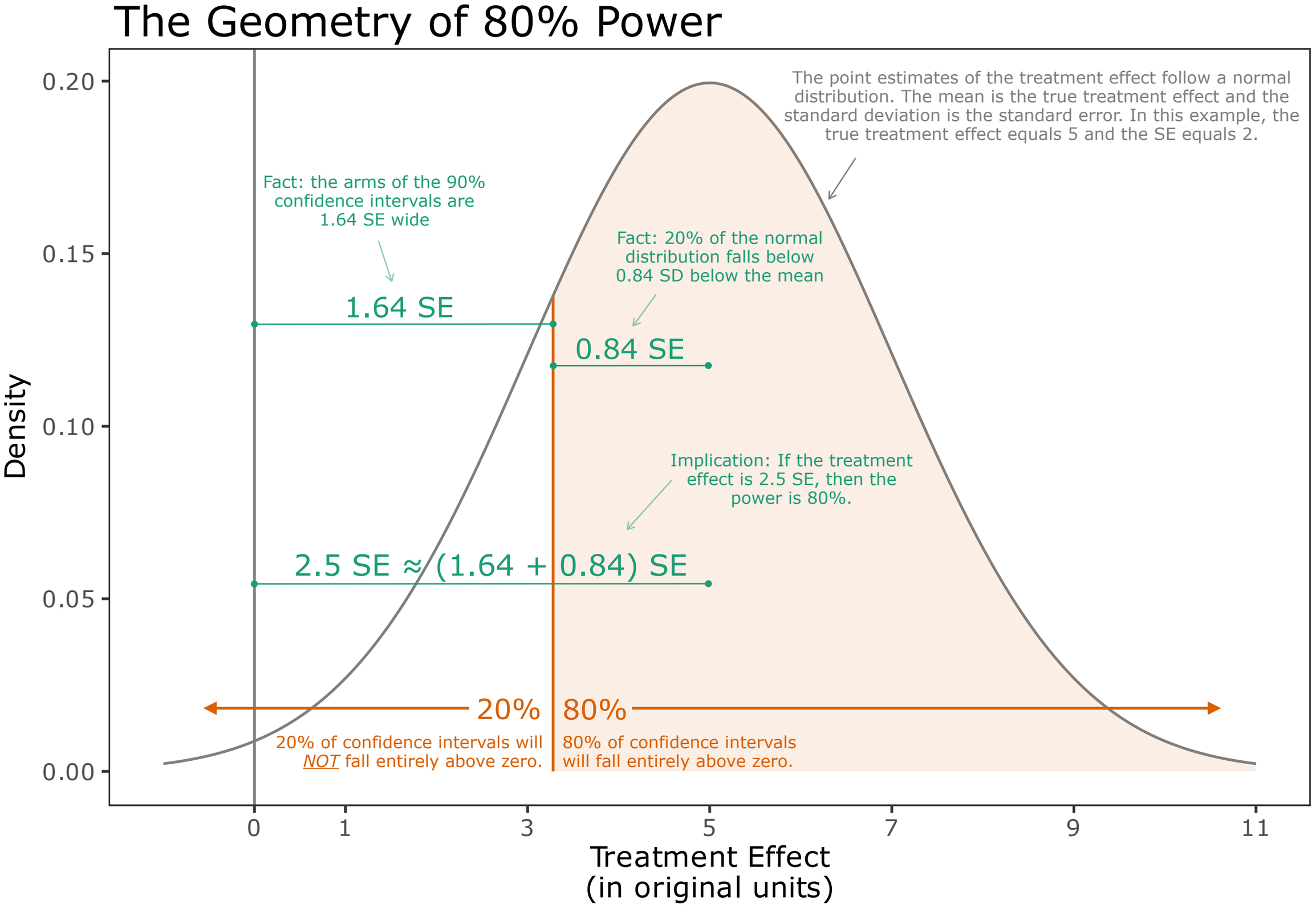
The Geometry of 80% Power
The point estimates of the treatment effect follow a normal distribution. The mean is the true treatment effect and the standard deviation is the standard error. In this example, the true treatment effect equals 5 and the SE equals 2.
Fact: the arms of the 90% confidence intervals are 1.64 SE wide
1.64 SE
Fact: 20% of the normal distribution falls below 0.84 SD below the mean
0.84 SE
Implication: If the treatment effect is 2.5 SE, then the power is 80%.
2.5 SE ≈ (1.64 + 0.84) SE
20%
80%
20% of confidence intervals will NOT fall entirely above zero.
80% of confidence intervals will fall entirely above zero.
Density
0.20
0.15
0.10
0.05
0.00
0
1
3
5
7
9
11
Treatment Effect (in original units)